

IPA ASSIGNMENT - 8

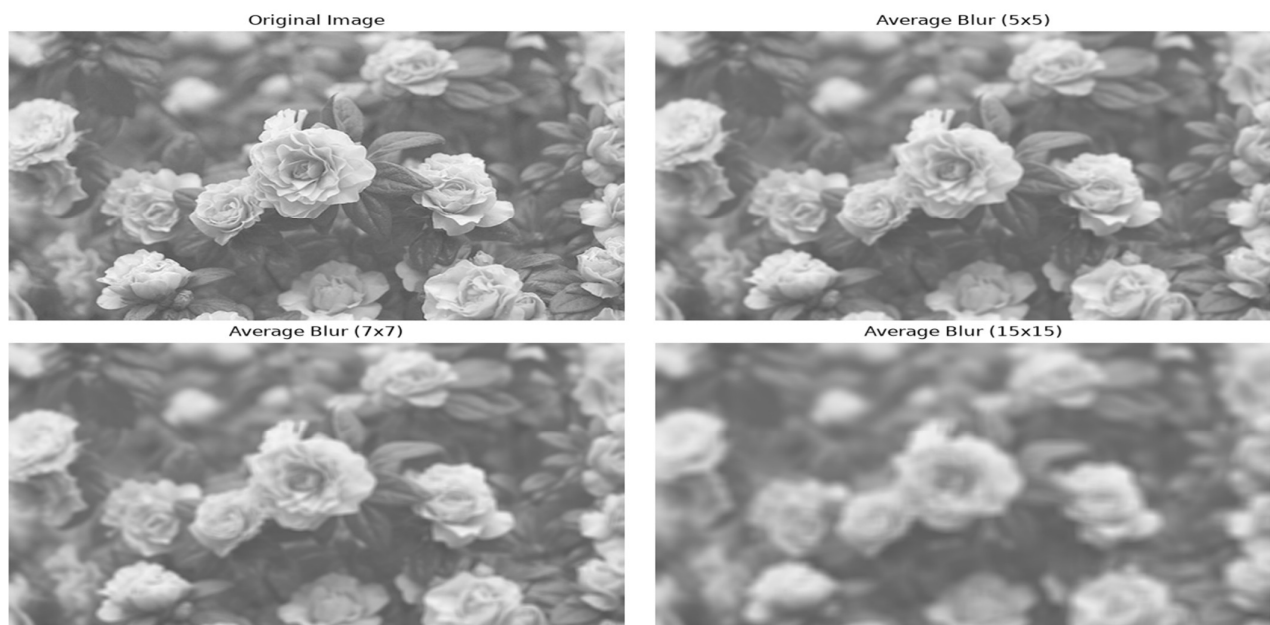
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1. Using a 5*5, 7*7 and 15*15 implement the standard average filter

```
import cv2
import matplotlib.pyplot as plt
image = cv2.imread("image_1.jpg")
if image is None: print("Error: Could not read image.") exit()
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
blur_5x5 = cv2.blur(image_rgb, (5, 5))
blur_7x7 = cv2.blur(image_rgb, (7, 7))
blur_15x15 = cv2.blur(image_rgb, (15, 15))
# Display all images together
plt.figure(figsize=(12, 8))
titles = [
    "Original Image",
    "Average Blur (5x5)",
    "Average Blur (7x7)",
    "Average Blur (15x15)", ]
images = [image_rgb, blur_5x5, blur_7x7, blur_15x15]
for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

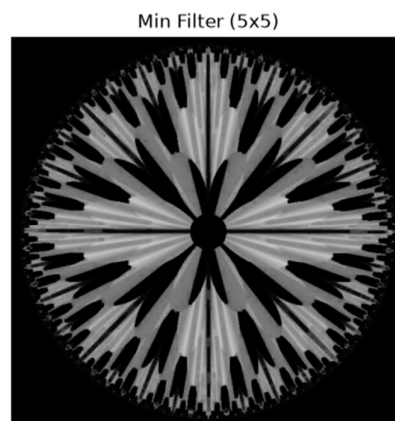
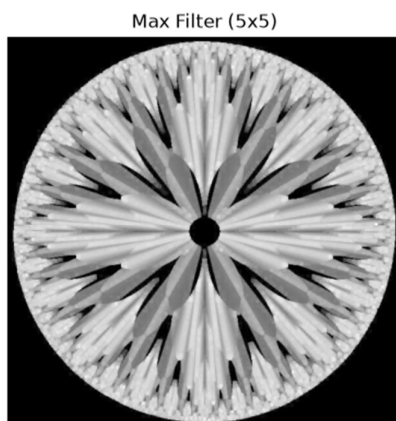
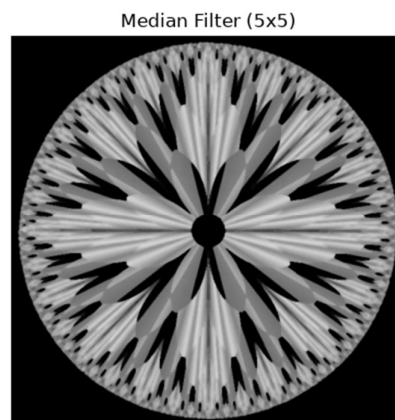
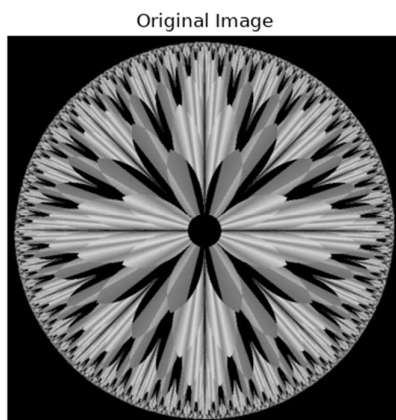
OUTPUT



2. Implement Median, Max and Min Filter

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("image.jpg")
if image is None:
    print("Error: Could not read image.")
    exit()
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
median = cv2.medianBlur(image_rgb, 5)
kernel = np.ones((5, 5), np.uint8)
max_filter = cv2.dilate(image_rgb, kernel)
min_filter = cv2.erode(image_rgb, kernel)
plt.figure(figsize=(12, 8))
titles = [
    "Original Image",
    "Median Filter (5x5)",
    "Max Filter (5x5)",
    "Min Filter (5x5)",
]
images = [image_rgb, median, max_filter, min_filter]
for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT



3. Laplacian Filter using the 3 Matrices

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("image_1.jpg", cv2.IMREAD_GRAYSCALE)
if image is None:
    print("Error: Could not read image.")
    exit()
kernel1 = np.array([[0, 1, 0], [1, -4, 1], [0, 1, 0]], dtype=np.float32)
kernel2 = np.array([[1, 1, 1], [1, -8, 1], [1, 1, 1]], dtype=np.float32)
kernel3 = np.array([[1, 0, 1], [0, -4, 0], [1, 0, 1]], dtype=np.float32)
lap1 = cv2.filter2D(image, cv2.CV_32F, kernel1)
lap2 = cv2.filter2D(image, cv2.CV_32F, kernel2)
lap3 = cv2.filter2D(image, cv2.CV_32F, kernel3)
lap1 = cv2.convertScaleAbs(lap1)
lap2 = cv2.convertScaleAbs(lap2)
lap3 = cv2.convertScaleAbs(lap3)
plt.figure(figsize=(12, 8))
titles = ["Original", "Laplacian Kernel 1", "Laplacian Kernel 2", "Laplacian Kernel 3"]
images = [image, lap1, lap2, lap3]
for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT

